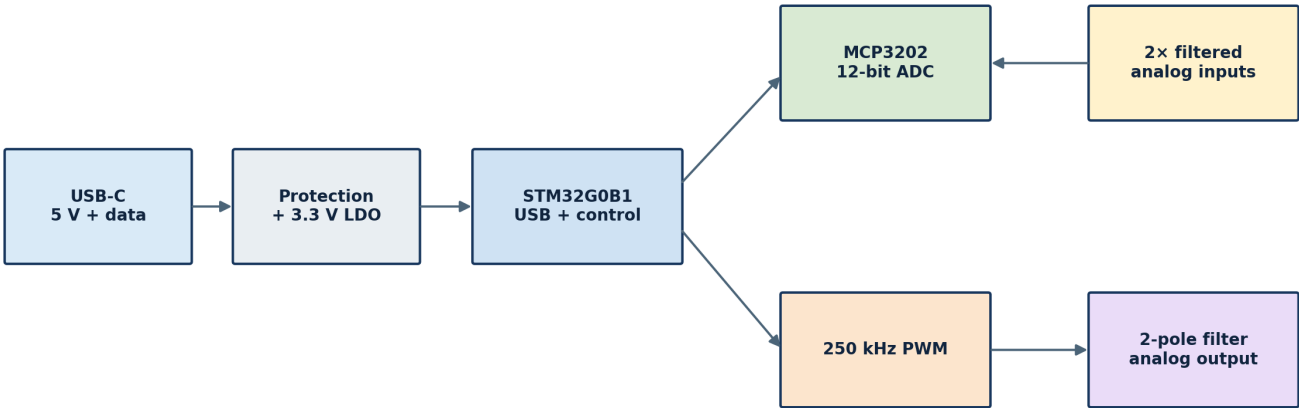


USB Mixed-Signal STM32 Data-Acquisition Board

A schematic-ready two-channel acquisition/control architecture connecting analog conditioning, external conversion, embedded buffering, USB transport, PCB constraints, and staged verification.

USB Mixed-Signal STM32 DAQ — Rev A Architecture



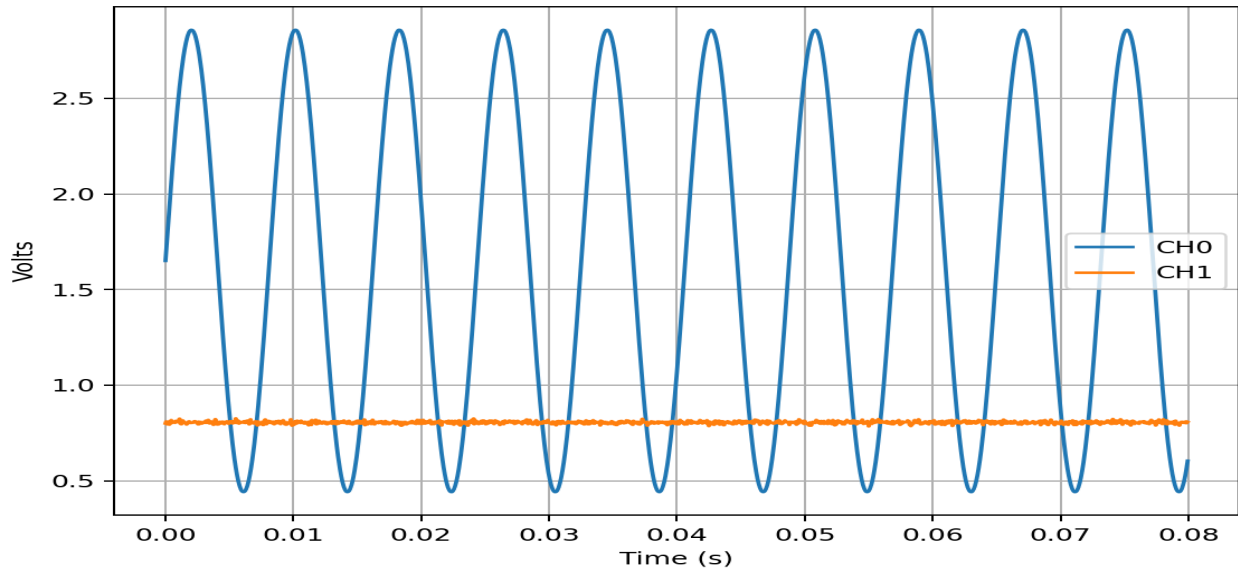
Requirements and architecture

The board targets two protected 0–3.0 V channels, 20 kS/s per channel, a 5 kHz useful signal band, 12-bit nominal conversion, USB CDC streaming, SWD, and a filtered PWM output. USB VBUS feeds a protected low-noise 3.3 V supply; a ferrite creates an analog supply zone while every ground pin connects to one continuous plane.

Block	Rev-A choice	Engineering reason
MCU	STM32G0B1CBT6	Native USB FS, SPI, timers, SWD, LQFP-48
ADC	MCP3202	Two-channel 12-bit SPI converter in an assembly-friendly package
Input	1 k Ω / 15 nF + TLV9062	10.61 kHz pole and buffering of the SAR input
Output	250 kHz PWM + two RC poles	Low-cost slow stimulus; not a precision DAC
Layout	Two layers, continuous ground	Placement-based partitioning without broken return paths

Firmware, host software, and verification

A timer-triggered acquisition path reads paired ADC samples into alternating 32-sample buffers. Frames include synchronization, sequence, timestamp, status, and CRC-16-CCITT so the host can reject corruption and identify gaps. USB transfer is kept outside the sampling callback. The Python host supports simulated or serial input, engineering-unit conversion, CSV export, and plotting.



Verified evidence

Executable calculations cover quantization, RC poles, ideal PWM attenuation, power, and data rate. Native C and Python tests cover portable framing, buffering, CRC, PWM conversion, parser behavior, and simulated capture. These checks do not establish STM32 HAL timing or electrical performance.

Required next gates

Exact symbol/package pin audit; CubeMX target configuration and ARM build; KiCad schematic and PCB capture; ERC, DRC, impedance geometry, 3D/Gerber review; assembly; current-limited power-up; SWD and USB enumeration; ADC DC/noise/sine testing; PWM ripple; loopback; and sustained streaming.

Verification boundary

No schematic capture, PCB layout, fabricated board, USB enumeration, or bench measurement is claimed. Rev A remains non-isolated and uncalibrated, uses the 3.3 V rail as ADC reference, and has a first-order input filter. The project is a schematic-ready design and software/calculation test package—not physically validated hardware.